

PVsyst - Simulation report

Grid-Connected System

Project: Solar_PV_5MWp_Thessaly

Variant: New simulation variant

Sheds, single array

System power: 5028 kWp

Kardítsa - Greece



Project: Solar_PV_5MWp_Thessaly

Variant: New simulation variant

PVsyst V8.1.5

VC1, Simulation date:

14/08/26 18:39

with V8.1.5

Project summary

Geographical Site

Kardí tsa

Greece

Situation

Latitude 39.3646 °(N)

Longitude 21.9238 °(E)

Altitude 102 m

Time zone UTC+2

Project settings

Albedo 0.20

Weather data

Kardí tsa

Meteonorm 9.0 dll, Sat=100% - Synthetic

System summary

Grid-Connected System

Orientation #1

Fixed plane

Tilt/Azimuth 30 / 0 °

System information

PV Array

Nb. of modules 8450 units

Pnom total 5028 kWp

Sheds, single array

Near Shadings

Linear shadings : Fast (table)

Inverters

Nb. of units 12.8 units

Total power 3850 kWac

Grid power limit 3500 kWac

Grid lim. Pnom ratio 1.437

User's needs

Unlimited load (grid)

Battery pack

Storage strategy: Peak shaving

Nb. of units 930 units

Voltage 768 V

Capacity 6250 Ah

Results summary

Simulation step Hourly

Produced Energy 7332917 kWh/year Specific production 1458 kWh/kWp/year Perf. Ratio PR 82.85 %

Table of contents

Project and results summary	2
General parameters, PV Array Characteristics, System losses	3
Near shading definition - Iso-shadings diagram	5
Main results	6
Loss diagram	7
Predef. graphs	8
Cost of the system	9
Financial analysis	11
CO ₂ Emission Balance	14

**PVsyst V8.1.5**

VC1, Simulation date:
14/08/26 18:39
with V8.1.5

General parameters**Grid-Connected System****Sheds, single array****Simulation step**

Hourly

Orientation #1**Fixed plane**

Tilt/Azimuth 30 / 0 °

Sheds configuration

Nb. of sheds 13 units

Single array

Shading limit angle

Limit profile angle 18.1 °

Sizes

Sheds spacing 11.0 m

Sensitive width 4.58 m

GCR Shading 41.8 %

Top inactive band 0.02 m

Bottom inactive band 0.02 m

Models used

Transposition Perez
Diffuse Perez, Meteonorm
Circumsolar separate

Horizon

Free Horizon

Near Shadings

Linear shadings : Fast (table)

Storage

Kind Peak shaving

User's needs

Unlimited load (grid)

Grid power limitation

Active power 3500 kWac

Pnom ratio 1.437

Limit applied at the injection point

PV Array Characteristics**PV module**

Manufacturer Generic
Model JKM-595N-72HL4-V
(Original PVsyst database)
Unit Nom. Power 595 Wp
Number of PV modules 8450 units
Nominal (STC) 5028 kWp
Modules 325 units x 26 In series
At operating cond. (50°C)
Pmpp 4663 kWp
U mpp 1056 V
I mpp 4417 A

Total PV power

Nominal (STC) 5028 kWp
Total 8450 modules
Module area 21828 m²
Cell area 20153 m²

Inverter

Manufacturer Generic
Model SUN2000-300KTL-H0
(Original PVsyst database)
Unit Nom. Power 300 kWac
Number of inverters 77 * MPPT 17% 12.8 units
Total power 3850 kWac
Operating voltage 550-1500 V
Max. power (=>30°C) 330 kWac
Pnom ratio (DC:AC) 1.31
No power sharing between MPPTs

Total inverter power

Total power 3850 kWac
Nb. of inverters 13 units
0.2 unused
Pnom ratio 1.31



PVsyst V8.1.5

VC1, Simulation date:
14/08/26 18:39
with V8.1.5

Grid storage - Peak shaving

Battery

Manufacturer Generic
Model Adjustable Li-Ion

Battery pack

Nb. of units 15 in series x 62 in parallel
Voltage 768 V
Nominal Capacity (C10) 6250 Ah
Nominal energy 4.80 MWh
Operating SOC limits 95.0 / 20.0 %
Useable energy 3.60 MWh
Operating temperature Fixed 20 °C

Battery input charger

Model Generic
Max. charging power 2.00 MWdc
Max./Euro effic. 98.0 / 96.5 %

Battery to Grid inverter

Model Generic
Max. discharging power 2.00 MWac
Max./Euro effic. 98.0 / 96.5 %

Array losses

Array Soiling Losses

Loss Fraction 1.5 %

Thermal Loss factor

Module temperature according to irradiance
Uc (const) 29.0 W/m²K
Uv (wind) 0.0 W/m²K/m/s

DC wiring losses

Global array res. 3.9 mΩ
Loss Fraction 1.50 % at STC

Module Quality Loss

Loss Fraction -0.75 %

Module mismatch losses

Loss Fraction 2.00 % at MPP

Strings Mismatch loss

Loss Fraction 0.10 %

IAM loss factor

Incidence effect (IAM): Fresnel, AR coating, n(glass)=1.526, n(AR)=1.290

0°	30°	50°	60°	70°	75°	80°	85°	90°
1.000	0.999	0.987	0.963	0.892	0.814	0.679	0.438	0.000

System losses

Unavailability of the system

Time fraction 2.0 %
7.3 days,
3 periods

AC wiring losses

Inv. output line up to MV transfo

Inverter voltage 800 Vac tri
Loss Fraction 1.00 % at STC

Inverter: SUN2000-300KTL-H0

Wire section (13 Inv.) Copper 13 x 3 x 150 mm²
Average wires length 132 m

AC losses in transformers

MV transfo

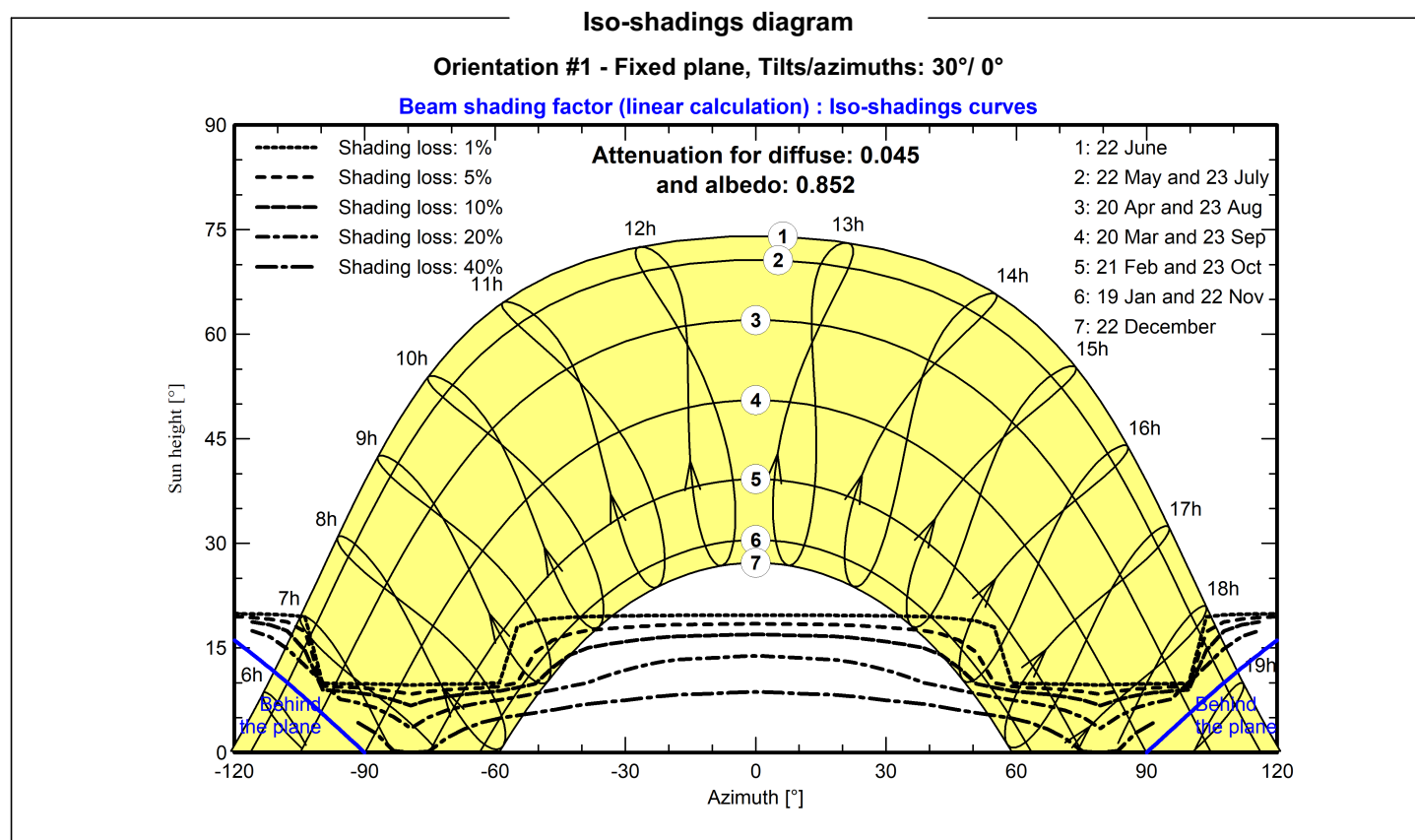
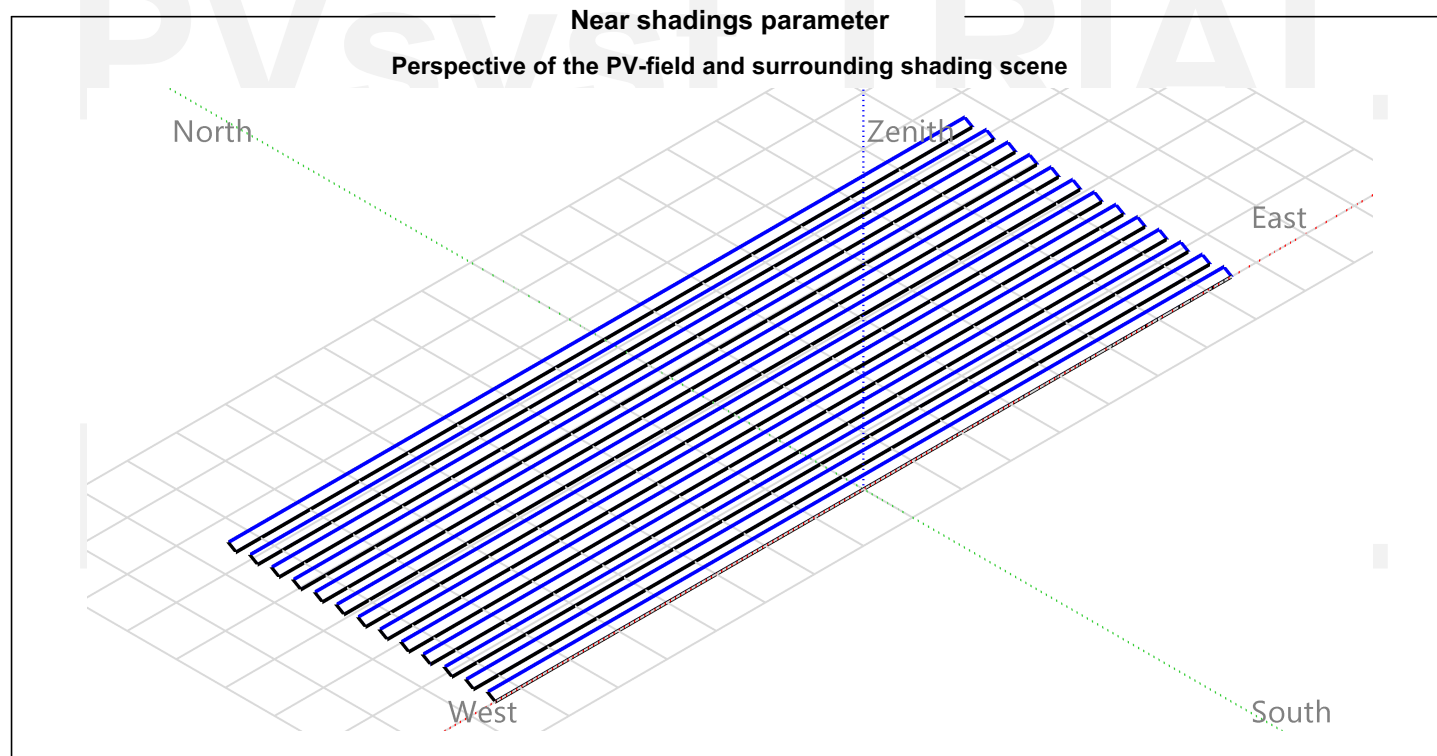
Medium voltage 20 kV

Transformer parameters

Nominal power at STC 4.98 MVA
Iron Loss (24/24 Connexion) 3.85 kVA
Iron loss fraction 0.08 % at STC
Copper loss 64.30 kVA
Copper loss fraction 1.29 % at STC
Coils equivalent resistance 3 x 1.66 mΩ

PVsyst V8.1.5

VC1, Simulation date:
14/08/26 18:39
with V8.1.5





PVsyst V8.1.5

VC1, Simulation date:

14/08/26 18:39

with V8.1.5

Main results

Simulation step

Hourly

System Production

Produced Energy 7332917 kWh/year

Specific production

1458 kWh/kWp/year

Perf. Ratio PR

82.85 %

Battery ageing (State of health)

Final battery state of health 99.8 %

Battery age 1. year

Economic evaluation

Investment

Global 4.044.500.00 EUR

Specific 0.80 EUR/Wp

Yearly cost

Annuities

282.053.52 EUR/yr

Run. costs

202.217.96 EUR/yr

Payback period

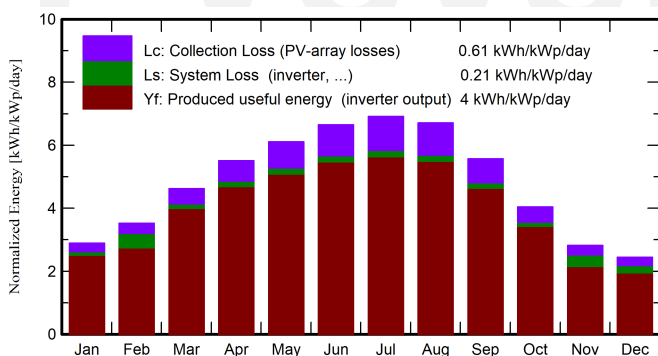
18.8 years

LCOE

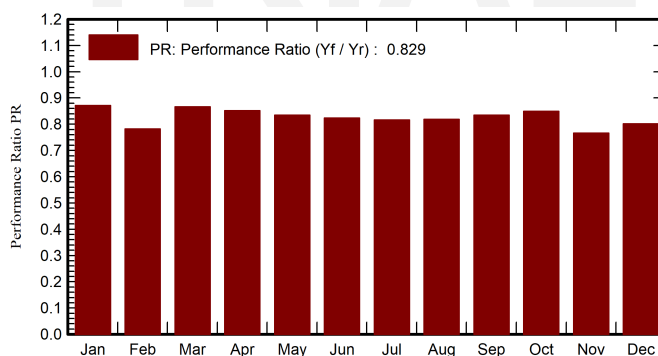
Energy cost

0.07 EUR/kWh

Normalized productions (per installed kWp)



Performance Ratio PR



Balances and main results

	GlobHor kWh/m ²	DiffHor kWh/m ²	T_Amb °C	GlobInc kWh/m ²	GlobEff kWh/m ²	EArray kWh	E_Grid kWh	EBatDis kWh	PR ratio
January	58.8	30.00	5.70	89.7	83.5	408922	392877	3197	0.871
February	72.1	35.32	7.33	98.6	93.3	452093	387655	6141	0.782
March	119.2	60.87	10.53	143.2	135.1	646536	623505	13243	0.866
April	153.9	72.56	13.97	165.2	155.8	734427	707775	20551	0.852
May	191.0	79.53	19.20	189.3	178.1	824434	793889	23364	0.834
June	209.9	82.50	24.00	199.5	187.5	856182	825612	19255	0.823
July	221.7	78.32	26.52	214.2	201.8	911403	878753	22291	0.816
August	198.1	71.33	26.12	208.0	196.7	887695	856452	23818	0.819
September	142.7	59.74	21.49	167.0	158.2	727147	700630	18995	0.835
October	96.4	45.91	16.44	125.3	118.4	555562	534667	12446	0.849
November	58.8	33.32	11.41	84.6	79.0	381609	325980	2379	0.767
December	48.0	26.48	6.87	75.7	69.4	341213	305122	1702	0.802
Year	1570.5	675.88	15.85	1760.3	1656.9	7727223	7332917	167383	0.829

Legends

GlobHor Global horizontal irradiation

DiffHor Horizontal diffuse irradiation

T_Amb Ambient Temperature

GlobInc Global incident in coll. plane

GlobEff Effective Global, corr. for IAM and shadings

EArray Effective energy at the output of the array

E_Grid Energy injected into grid

EBatDis Battery Discharging Energy

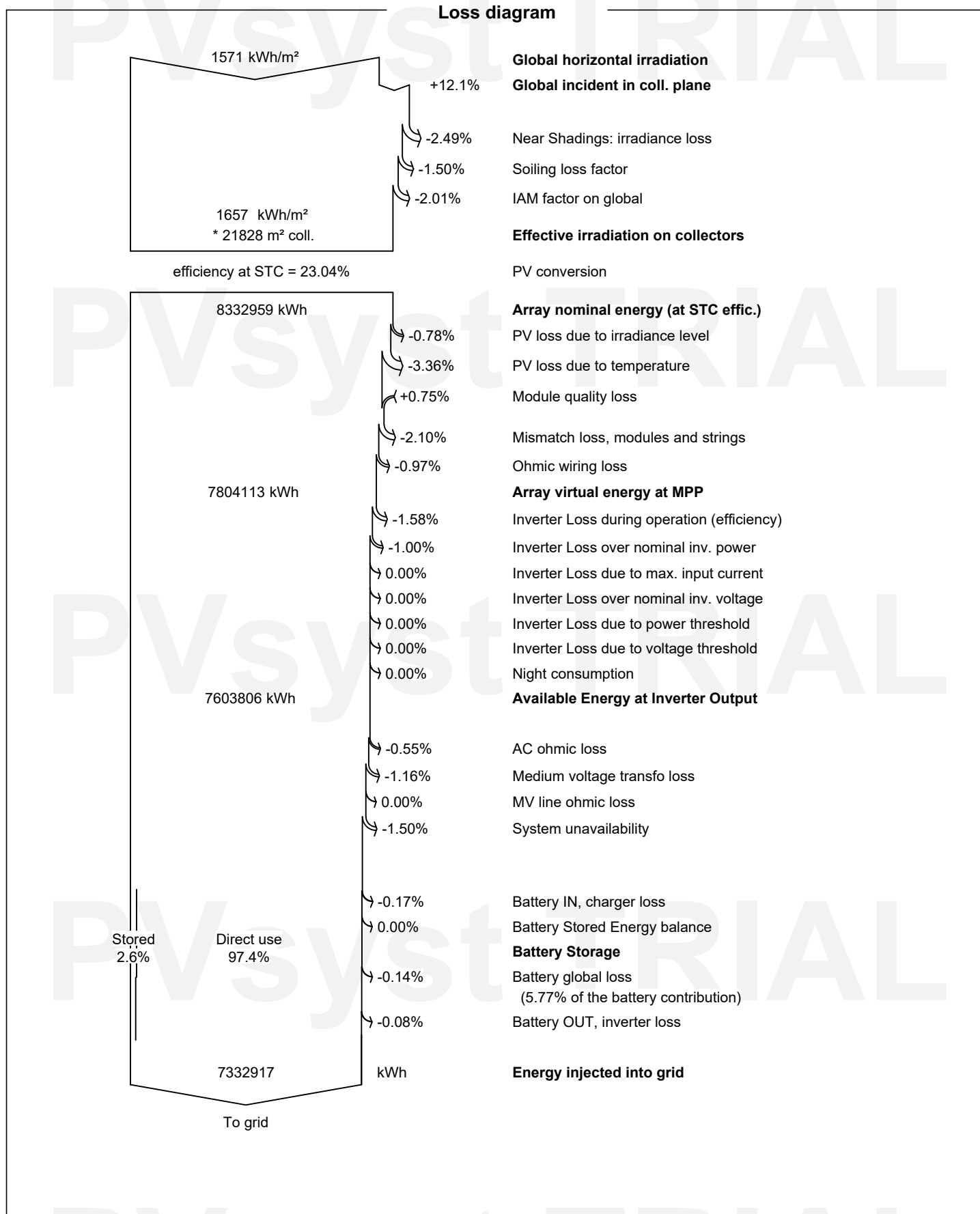
PR Performance Ratio

PVsyst V8.1.5

VC1, Simulation date:

14/08/26 18:39

with V8.1.5



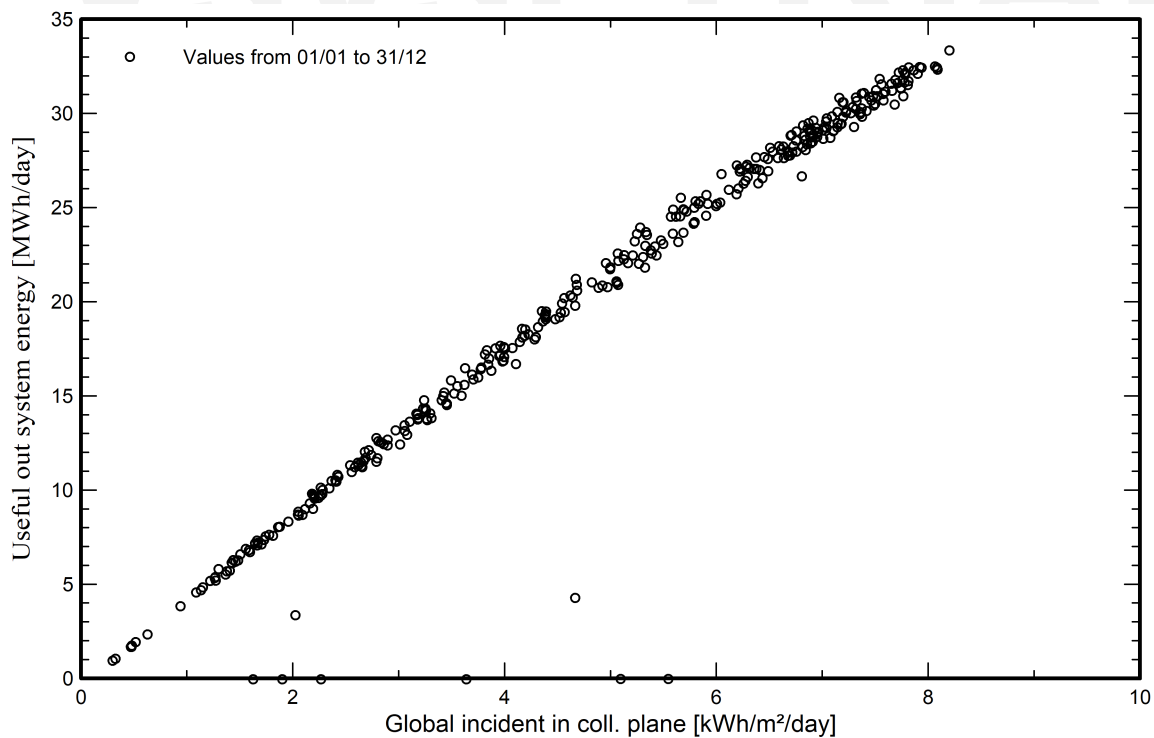


PVsyst V8.1.5

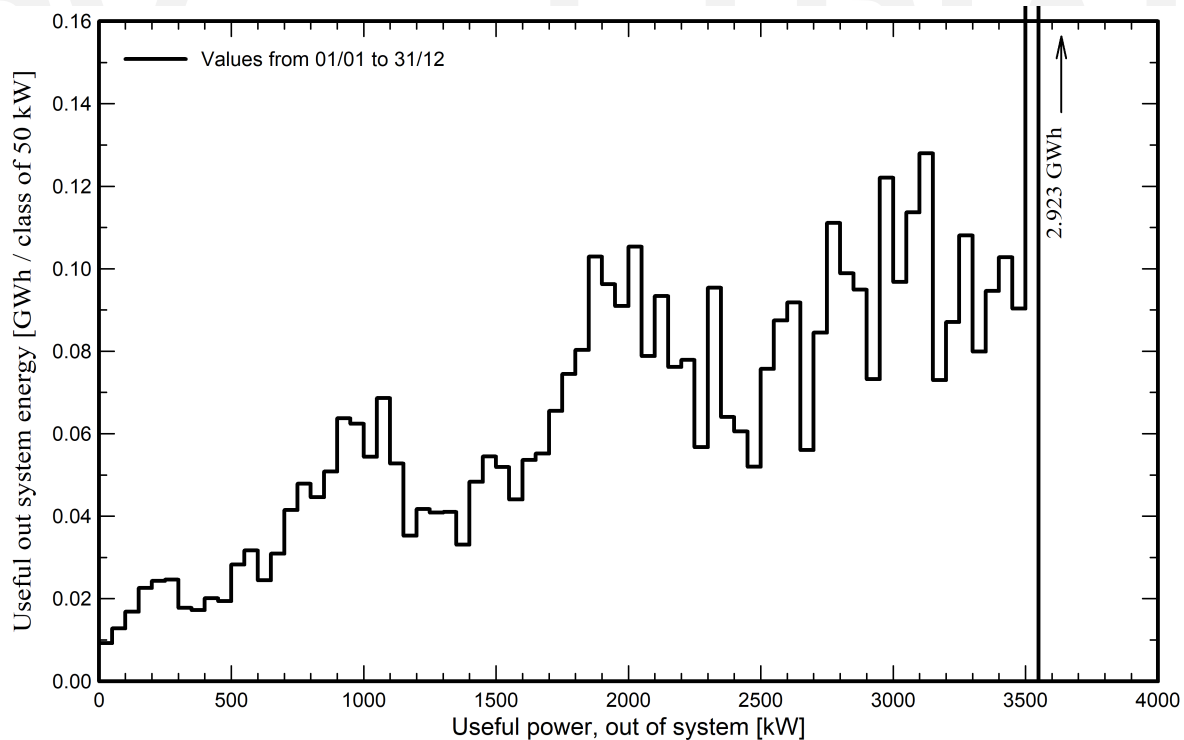
VC1, Simulation date:
14/08/26 18:39
with V8.1.5

Predef. graphs

Daily Input/Output diagram



System Output Power Distribution





PVsyst V8.1.5

VC1, Simulation date:

14/08/26 18:39

with V8.1.5

Cost of the system

Installation costs

Item	Quantity units	Cost EUR	Total EUR
PV modules			
JKM-595N-72HL4-V	8450	75.00	633.750.00
Supports for modules	8450	35.00	295.750.00
Inverters			
SUN2000-300KTL-H0	13	15.000.00	192.500.00
Batteries	4800	230.00	1.104.000.00
Other components			
Accessories, fasteners	1	800.000.00	800.000.00
Wiring	1	250.000.00	250.000.00
Monitoring system, display screen	1	50.000.00	50.000.00
Studies and analysis			
Engineering	1	60.000.00	60.000.00
Permits and other adm. Fees	1	40.000.00	40.000.00
Environmental studies	1	15.000.00	15.000.00
Economic analysis	1	5.000.00	5.000.00
Installation			
Global installation cost per module	8450	30.00	253.500.00
Transport	1	40.000.00	40.000.00
Settings	1	20.000.00	20.000.00
Grid connection	1	200.000.00	200.000.00
Insurance			
Building insurance	1	15.000.00	15.000.00
Transport insurance	1	5.000.00	5.000.00
Liability insurance	1	5.000.00	5.000.00
Land costs			
Land preparation	1	30.000.00	30.000.00
Loan bank charges			30.000.00
		Total	4.044.500.00
		Depreciable asset	3.026.000.00

Operating costs

Item	Amount EUR	Payment schedule
Maintenance		
Inverter replacement	12.833.33	Every year
Repairs	10.000.00	Every year
Cleaning	5.000.00	Every year
Battery replacement	92.000.00	Every year
Security fund	3.000.00	Every year
Land rent	15.000.00	Every year
Insurance		
Facilities insurance	7.000.00	Every year
Liability insurance	1.500.00	Every year
Business interruption insurance	1.500.00	Every year
Administrative, accounting	10.000.00	Every year
Yearly average operating cost (OPEX)	157.833.33	
Including inflation (2.00%)	202.217.96	



PVsyst V8.1.5

VC1, Simulation date:
14/08/26 18:39
with V8.1.5

Cost of the system

System summary

Total installation cost	4.044.500.00 EUR
Operating costs (incl. inflation 2.00%/year)	202.217.96 EUR/year
Produced Energy	7350 MWh/year
Cost of produced energy (LCOE)	0.0705 EUR/kWh

**PVsyst V8.1.5**

VC1, Simulation date:

14/08/26 18:39

with V8.1.5

Financial analysis**Simulation period**

Project lifetime 25 years Start year 2027

Income variation over time

Inflation 2.00 %/year

Module Degradation 0.40 %/year

Discount rate 6.00 %/year

Income dependent expenses

Income tax rate 22.00 %/year

Other income tax 0.00 %/year

Dividends 5.00 %/year

Depreciable assets

Asset	Depreciation method	Depreciation period (years)	Salvage value (EUR)	Depreciable (EUR)
PV modules				
JKM-595N-72HL4-V	Straight-line	20	0.00	633.750.00
Supports for modules	Straight-line	20	0.00	295.750.00
Inverters				
SUN2000-300KTL-H0	Straight-line	20	0.00	192.500.00
Batteries	Straight-line	20	0.00	1.104.000.00
Accessories, fasteners	Straight-line	20	0.00	800.000.00
		Total	0.00	3.026.000.00

Financing

Own funds 1.213.365.00 EUR

Loan - Redeemable with fixed annuity - 15 years 2.831.135.00 EUR

Interest rate: 5.50%/year

Electricity sale

Feed-in tariff 0.08000 EUR/kWh

Duration of tariff warranty 20 years

Annual connection tax 0.00 EUR/year

Annual tariff variation 0.0 %/year

Feed-in tariff decrease after warranty 0.00 %

Return on investment

Payback period 18.8 years

Net present value (NPV) 405.404.43 EUR

Internal rate of return (IRR) 8.61 %

Return on investment (ROI) 10.0 %

Paid dividends 186.569.97 EUR



PVsyst V8.1.5

VC1, Simulation date:

14/08/26 18:39

with V8.1.5

Financial analysis

Detailed economic results (EUR)

Year	Electricity sale	Own funds	Loan principal	Loan interest	Run. costs	Deprec. allow.	Taxable income	Taxes	After-tax profit	Divid. 5.00%	Cumul. profit	% amorti.
0	0	1.213.365	0	0	0	0	0	0	0	0	-1.213.365	0.0%
1	587.982	0	126.341	155.712	157.833	151.300	123.136	27.090	121.005	6.050	-1.099.209	5.9%
2	585.630	0	133.290	148.764	160.990	151.300	124.576	27.407	115.180	5.759	-996.700	11.8%
3	583.287	0	140.621	141.433	164.210	151.300	126.345	27.796	109.228	5.461	-904.990	17.5%
4	580.954	0	148.355	133.699	167.494	151.300	128.462	28.262	103.145	5.157	-823.289	23.2%
5	578.630	0	156.514	125.539	170.844	151.300	130.947	28.808	96.925	4.846	-750.862	28.9%
6	576.316	0	165.123	116.931	174.261	151.300	133.824	29.441	90.560	4.528	-687.020	34.5%
7	574.011	0	174.205	107.849	177.746	151.300	137.116	30.165	84.046	4.202	-631.125	40.2%
8	571.715	0	183.786	98.268	181.301	151.300	140.846	30.986	77.374	3.869	-582.580	46.0%
9	569.428	0	193.894	88.160	184.927	151.300	145.041	31.909	70.538	3.527	-540.828	51.8%
10	567.150	0	204.558	77.495	188.625	151.300	149.729	32.940	63.531	3.177	-505.353	57.7%
11	564.881	0	215.809	66.245	192.398	151.300	154.939	34.087	56.343	2.817	-475.672	63.8%
12	562.622	0	227.678	54.375	196.246	151.300	160.701	35.354	48.968	2.448	-451.337	70.0%
13	560.371	0	240.201	41.853	200.171	151.300	167.048	36.750	41.397	2.070	-431.928	76.4%
14	558.130	0	253.412	28.642	204.174	151.300	174.014	38.283	33.619	1.681	-417.058	83.1%
15	555.897	0	267.349	14.704	208.258	151.300	181.635	39.960	25.626	1.281	-406.366	90.0%
16	553.674	0	0	0	212.423	151.300	189.951	41.789	299.462	14.973	-288.484	92.9%
17	551.459	0	0	0	216.671	151.300	183.488	40.367	294.420	14.721	-179.146	95.6%
18	549.253	0	0	0	221.005	151.300	176.948	38.929	289.320	14.466	-77.785	98.1%
19	547.056	0	0	0	225.425	151.300	170.331	37.473	284.158	14.208	16.133	100.4%
20	544.868	0	0	0	229.933	151.300	163.635	36.000	278.935	13.947	103.106	102.5%
21	542.688	0	0	0	234.532	0	308.156	67.794	240.362	12.018	173.810	104.3%
22	540.518	0	0	0	239.223	0	301.295	66.285	235.010	11.751	239.027	105.9%
23	538.356	0	0	0	244.007	0	294.349	64.757	229.592	11.480	299.133	107.4%
24	536.202	0	0	0	248.887	0	287.315	63.209	224.106	11.205	354.483	108.8%
25	534.057	0	0	0	253.865	0	280.192	61.642	218.550	10.928	405.404	110.0%
Total	14.015.135	1.213.365	2.831.135	1.399.668	5.055.449	3.026.000	4.534.019	997.484	3.731.399	186.570	405.404	110.0%

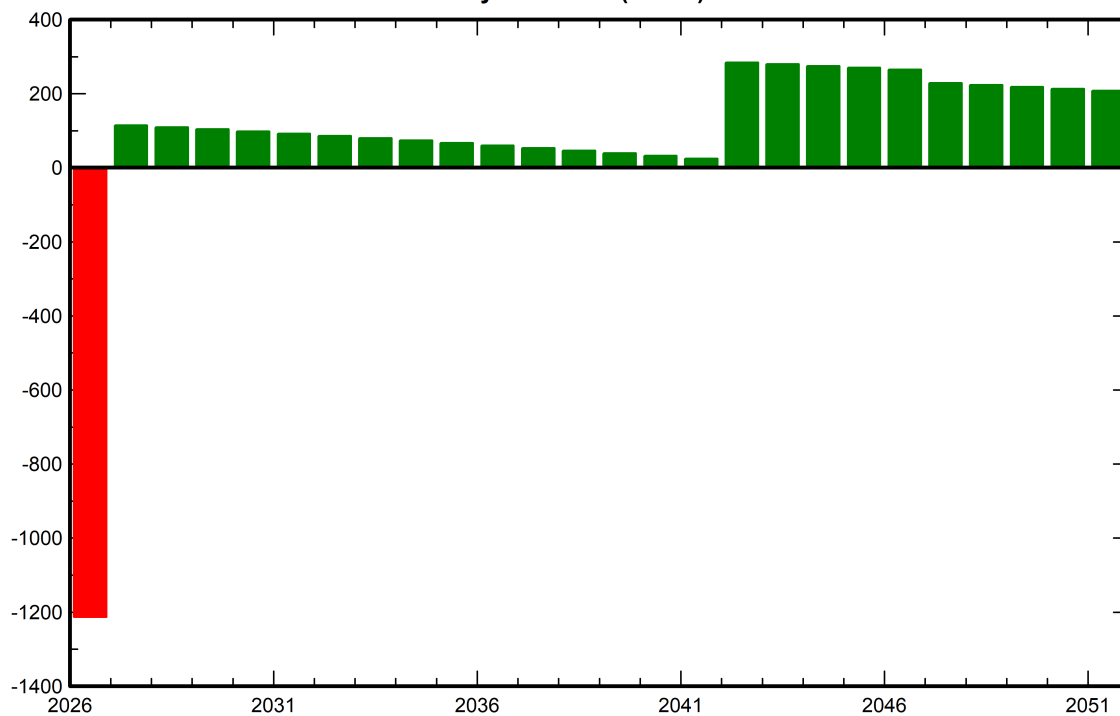


PVsyst V8.1.5

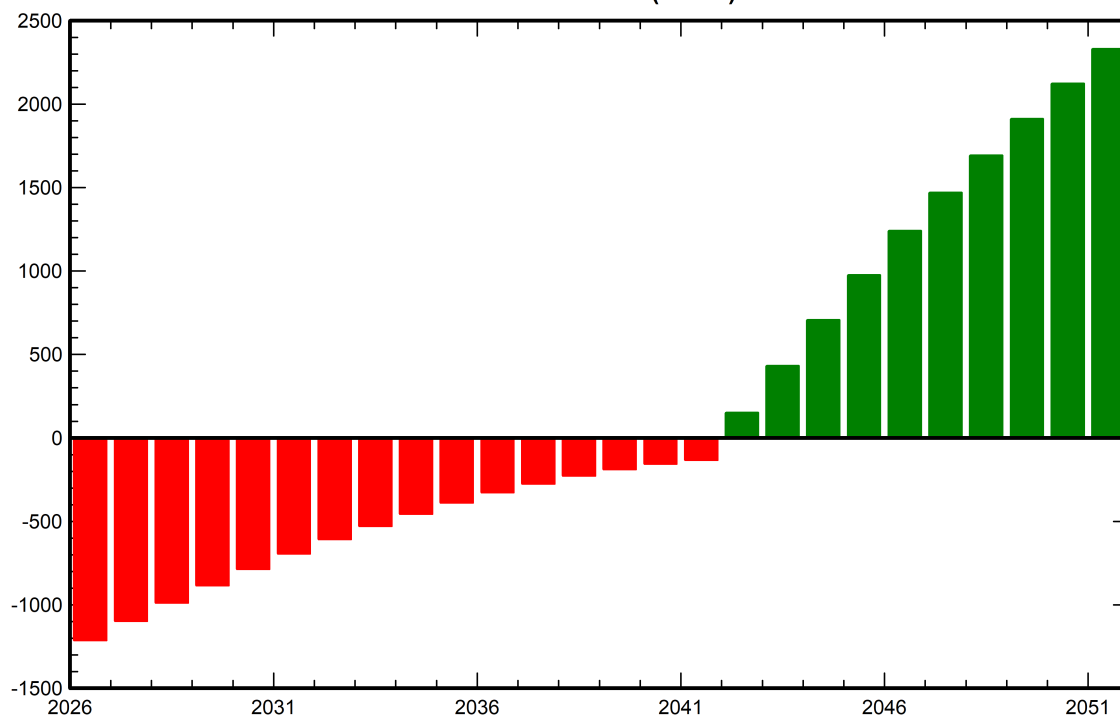
VC1, Simulation date:
14/08/26 18:39
with V8.1.5

Financial analysis

Yearly cashflow (kEUR)



Cumulative cashflow (kEUR)





PVsyst V8.1.5

VC1, Simulation date:
14/08/26 18:39
with V8.1.5

CO₂ Emission Balance

Total: 130850.3 tCO₂

Generated emissions

Total: 9028.84 tCO₂

Source: Detailed calculation from table below

Replaced Emissions

Total: 161213.3 tCO₂

System production: 7361.34 MWh/yr

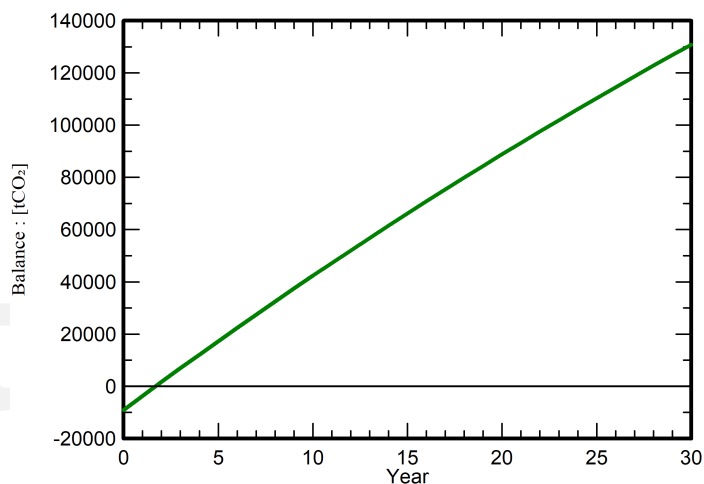
Grid Lifecycle Emissions: 730 gCO₂/kWh

Source: IEA List

Country: Greece

Lifetime: 30 years

Annual degradation: 1.0 %

Saved CO₂ Emission vs. Time

System Lifecycle Emissions Details

Item	LCE	Quantity	Subtotal
			[kgCO ₂]
Modules	1713 kgCO ₂ /kWp	5028 kWp	8611128
Supports	4.87 kgCO ₂ /kg	84500 kg	411439
Inverters	482 kgCO ₂ /units	13.0 units	6271